

# A natural proof of Quadratic Reciprocity

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## 1 Introduction

The purpose of this text is to give a proof of Quadratic Reciprocity which is, in some sense, an *additive version of the standard proof of Gauss's Lemma*.<sup>1</sup> Here is a more precise formulation:

A common proof of Quadratic Reciprocity proceeds in three steps:

1. Prove Gauss's Lemma.
2. Derive Corollary 5 below from Gauss's Lemma.
3. Derive Quadratic Reciprocity from Corollary 5.

This is done in Hardy and Wright [1], in Niven, Zuckerman and Montgomery [2], and in Wallach [3] (and probably in many other texts). Steps 1 and 3 are fairly standard, but there are numerous ways of doing Step 2. Our main observation is that a minor variation of the standard argument used in Step 1 works for Step 2.

To elaborate: the assumptions of Gauss's Lemma and of Theorem 2 are that  $p$  is an odd prime and  $a$  an integer prime to  $p$ . The standard argument in question goes as follows: we set

$$I = \{1, 2, \dots, \frac{p-1}{2}\}, \quad U = \{a, 2a, 3a, \dots, \frac{p-1}{2}a\}, \quad (1)$$

we define two other sets  $M$  and  $V$ ,<sup>2</sup> we prove  $M = I$  and  $V = U$ , and we derive the conclusion from the equalities

$$\prod_{i \in I} i = \prod_{m \in M} m \quad \text{and} \quad \prod_{u \in U} u = \prod_{v \in V} v. \quad (2)$$

Our point is that Step 2 can be easily accomplished via the additive analog

$$\sum_{i \in I} i = \sum_{m \in M} m \quad \text{and} \quad \sum_{u \in U} u = \sum_{v \in V} v \quad (3)$$

of the equalities (2).

Step 3 is basically the same in [1], [2] and [3], and also in this text. Corollary 5 is a corollary to Theorem 2. Theorem 2 is implicit in the proof of Theorem 3.3 in [2]; it is even more implicit in [1] (Section 6.13); it appears as Lemma 10 in [3]. Theorem 2 yields the most transparent proof I know of the second supplement<sup>3</sup> to the law of Quadratic Reciprocity (this argument was already given in [2]—see again the proof of Theorem 3.3 there).

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<sup>1</sup>This text is available at <https://github.com/Pierre-Yves-Gaillard/quadratic-reciprocity>

<sup>2</sup>The sets  $M$  and  $V$  are respectively defined in (5) and (7) below.

<sup>3</sup>This is the equality  $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$ .

In the last two sections we give two proofs of Gauss's and a generalization of Euler's Criterion.

Until further notice,  $p$  and  $q$  are distinct odd primes, and  $a$  is an integer prime to  $p$ . Recall that the **Legendre symbol**  $(\frac{a}{p})$  is equal to 1 if  $a$  is a square  $(\bmod p)$ , and to  $-1$  if  $a$  is a non-square  $(\bmod p)$ . (Since the integers  $x$  and  $y$  have the same square  $(\bmod p)$  if and only if  $y \equiv \pm x \pmod{p}$ , we see that there are  $(p-1)/2$  nonzero square residues, and  $(p-1)/2$  non-square residues  $(\bmod p)$ .) Our main goal is to prove Quadratic Reciprocity:

**Theorem 1** (Quadratic Reciprocity). *If  $p$  and  $q$  are distinct odd primes, then we have  $(\frac{p}{q})(\frac{q}{p}) = (-1)^{(p-1)(q-1)/4}$ .*

## 2 Proof of Gauss's Lemma and Theorem 2

Let  $\lfloor x \rfloor$  denote the floor of the real number  $x$  (the greatest integer less than or equal to  $x$ ), and set

$$f(a, p) = \sum_{i=1}^{(p-1)/2} \left\lfloor \frac{ia}{p} \right\rfloor.$$

Quadratic Reciprocity will follow from the theorem below:

**Theorem 2.** *If  $p$  is an odd prime and  $a$  an integer prime to  $p$ , then we have  $(\frac{a}{p}) = (-1)^\ell$  with*

$$\ell = (a+1) \frac{p^2-1}{8} + f(a, p).$$

**Remark 3.** *We have*

$$\frac{p^2-1}{8} = \sum_{i=1}^{(p-1)/2} i;$$

*in particular  $(p^2-1)/8$  is an integer. (One can also observe that  $\frac{p-1}{2} \frac{p+1}{2}$  is even.)*

The proof will proceed by comparing multiplicative and additive analogs of the same combinatorial identities.

**Corollary 4.** *We have  $(\frac{2}{p}) = (-1)^{(p^2-1)/8}$ .*

This follows from the fact that  $f(2, p) = 0$ , because, for  $1 \leq i \leq (p-1)/2$ , we have  $2i \leq p-1 < p$ , so  $\lfloor 2i/p \rfloor = 0$ .

**Corollary 5.** *We have  $(\frac{a}{p}) = (-1)^{f(a, p)}$  if  $a$  is odd.*

To prove Theorem 2 we shall use Euler's Criterion:

**Theorem 6** (Euler's Criterion). *Recall that  $p$  is an odd prime and  $a$  is an integer prime to  $p$ . Set  $n = (p-1)/2$ . Then we have  $(\frac{a}{p}) \equiv a^n \pmod{p}$ .*

A generalization of Euler's Criterion is given in the last section.

**Corollary 7.** *We have  $(\frac{ab}{p}) = (\frac{a}{p})(\frac{b}{p})$  if  $a$  and  $b$  are integers prime to  $p$ . We also have  $(\frac{1}{p}) = 1$ .*

To prove Euler's Criterion we use the following theorem:

**Theorem 8.** *The multiplicative group  $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$  of the field  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$  is cyclic (of order  $p - 1$ ).*

For a proof of Theorem 8, we refer the reader to Weil's book [4], Theorem X.1. A generalization of Euler's Criterion is given in the last section.

*Proof of Euler's Criterion (Theorem 6).* If  $b$  is in  $\mathbb{F}_p^*$ , we have  $b^n = \pm 1$  (because  $(b^n)^2 = b^{2n} = 1$ ) and we must show that  $b^n = 1$  if and only if  $b$  is a square. Let  $\zeta$  be a generator of  $\mathbb{F}_p^*$ . Then  $b = \zeta^j$  for some integer  $j$ . Then we get  $b^n = \zeta^{jn}$ , and  $b^n = 1$  if and only if  $2n$  divides  $jn$ , which means  $j$  is even; this, in turn, means  $b$  is a square.  $\square$

For  $i \in I$  (see (1)), let  $r_i$  be the remainder when  $ia$  is divided by  $p$ , so that we have

$$ia = p \left\lfloor \frac{ia}{p} \right\rfloor + r_i. \quad (4)$$

In particular, the  $r_i$  are distinct and satisfy  $1 \leq r_i \leq p - 1$ . Set  $J = \{i \in I \mid r_i \leq n\}$ ,  $K = \{i \in I \mid r_i \geq n + 1\}$ . We denote the cardinality of the set  $X$  by  $|X|$  and we indicate by  $X = Y \sqcup Z$  the fact that  $X$  is the disjoint union of  $Y$  and  $Z$ .

**Theorem 9** (Gauss's Lemma). *We have  $\left(\frac{a}{p}\right) = (-1)^{|K|}$ .*

*Proof of Gauss's Lemma (Theorem 9) and of Theorem 2.* The equality  $I = J \sqcup K$  is clear. Set

$$M_1 = \{r_j \mid j \in J\}, \quad M_2 = \{p - r_k \mid k \in K\}, \quad M = M_1 \cup M_2. \quad (5)$$

We claim

$$I = M_1 \sqcup M_2 = M. \quad (6)$$

To prove (6), note that, since the  $r_i$  are distinct (indeed if  $r_i = r_j$  then  $p \mid a(i - j)$ , and since  $\gcd(a, p) = 1$  and  $|i - j| < p$  we get  $i = j$ ), it suffices to show  $r_j \neq p - r_k$  for  $j \in J, k \in K$ . But  $r_j = p - r_k$  would imply  $0 \equiv r_j + r_k \equiv a(j + k) \pmod{p}$ , hence  $j + k \equiv 0 \pmod{p}$ , hence  $p$  divides  $j + k$ , contradicting the inequalities  $2 \leq j + k \leq p - 1$ . This proves (6).

The sets  $I, U, M$  and  $V$  in (2) and (3) are defined by (1), (5) and

$$V = \left\{ p \left\lfloor \frac{ia}{p} \right\rfloor + r_i \mid i \in I \right\}. \quad (7)$$

The equalities  $M = I$  and  $V = U$  follow respectively from (6) and (4).

In the remainder of this section  $i, j$  and  $k$  run respectively over  $I, J$  and  $K$ .

In view of (1), (5) and (7), the first and second equalities in (2) give respectively

$$n! = \prod i = \left( \prod r_j \right) \left( \prod (p - r_k) \right) \equiv (-1)^{|K|} \prod r_i \pmod{p} \quad (8)$$

and

$$n! a^n = \prod ia = \prod \left( p \left\lfloor \frac{ia}{p} \right\rfloor + r_i \right) \equiv \prod r_i \pmod{p}. \quad (9)$$

Similarly, the first and second equalities in (3) give respectively

$$\frac{p^2 - 1}{8} = \sum i = \sum r_j + \sum (p - r_k) = \sum r_j + p|K| - \sum r_k \equiv \sum r_i + |K| \pmod{2} \quad (10)$$

and

$$a \frac{p^2 - 1}{8} = \sum_{i=1}^n ia = pf(a, p) + \sum r_i \equiv f(a, p) + \sum r_i \pmod{2}. \quad (11)$$

Now Gauss's Lemma (Theorem 9) follows from Euler's Criterion (Theorem 6), (8) and (9). To prove Theorem 2, it suffices to show  $|K| \equiv \ell \pmod{2}$  (in the notation of Theorems 9 and 2). We have

$$|K| \equiv \frac{p^2 - 1}{8} + \sum r_i \equiv \frac{p^2 - 1}{8} + a \frac{p^2 - 1}{8} + f(a, p) \equiv \ell \pmod{2},$$

the first congruence following from (10) and the second from (11).  $\square$

### 3 Proof of Quadratic Reciprocity

We prove Quadratic Reciprocity (Theorem 1). Recall that  $p$  and  $q$  are distinct odd primes. By Corollary 5, it suffices to show

$$f(p, q) + f(q, p) = \frac{(p-1)(q-1)}{4}.$$

For any subset  $S$  of  $\mathbb{R}^2$ , put  $S^* = S \cap \mathbb{Z}^2$ . Consider the open rectangle

$$R = \{(x, y) \in \mathbb{R}^2 \mid 0 < x < \frac{p}{2}, 0 < y < \frac{q}{2}\},$$

the diagonal

$$D = \{(x, y) \in R \mid py = qx\},$$

the semi-open triangle above the diagonal

$$A = \{(x, y) \in R \mid py \geq qx\},$$

and the semi-open triangle below the diagonal

$$B = \{(x, y) \in R \mid py \leq qx\}.$$

Recall that, if  $X, Y, Z$  are sets, we denote the cardinality of  $X$  by  $|X|$  and we indicate by  $X = Y \sqcup Z$  the fact that  $X$  is the disjoint union of  $Y$  and  $Z$ . We have  $D^* = \emptyset$ . Indeed,  $py = qx$  with  $(x, y) \in R^*$  would imply that  $p$  divides  $x$  (because  $p$  and  $q$  are relatively prime), which is impossible since  $0 < x < p/2$ . We get

$$A^* \cap B^* = D^* = \emptyset, \quad R^* = A^* \sqcup B^*, \quad |A^*| + |B^*| = |R^*| = \frac{(p-1)(q-1)}{4}.$$

From the equations displayed above, it is enough to show  $f(p, q) = |A^*|$  and  $f(q, p) = |B^*|$ . Let us prove  $f(q, p) = |B^*|$ . Recall the definition of Iverson's bracket: if  $(S)$  is a statement, then  $[(S)] = 1$  if  $(S)$  is true, and  $[(S)] = 0$  if  $(S)$  is false. Recall also that, if  $t \geq 0$ , then  $[t]$  is the number of positive integers  $\leq t$ :

$$[t] = \sum_{i \in \mathbb{Z}} [1 \leq i \leq t] \quad \text{for } t \geq 0.$$

We have

$$|B^*| = \sum_{i=1}^{(p-1)/2} \sum_{j \in \mathbb{Z}} \left[ 1 \leq j \leq \frac{iq}{p} \right] = \sum_{i=1}^{(p-1)/2} \left[ \frac{iq}{p} \right] = f(q, p).$$

The equality  $f(p, q) = |A^*|$  is proved similarly. This proves Quadratic Reciprocity.

## 4 Two proofs of Gauss's Lemma

We add two proofs of Gauss's Lemma (or, more precisely, of a version of Gauss's Lemma which is slightly stronger than the one in Theorem 9).

Recall that  $p$  is an odd prime, that  $n = (p-1)/2$ , that  $\mathbb{F}_p$  is the field  $\mathbb{Z}/p\mathbb{Z}$ , and that  $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$  is its multiplicative group (see Theorem 8). Let  $a \in \mathbb{F}_p^*$  and let  $x_1, \dots, x_n \in \mathbb{F}_p^*$  satisfy

$$\mathbb{F}_p^* = \{x_1, -x_1, \dots, x_n, -x_n\}.$$

There is a unique permutation  $\sigma$  of  $I = \{1, \dots, n\}$  such that  $ax_i = \pm x_{\sigma(i)}$  for all  $i \in I$ . Recall that  $(\frac{a}{p})$  is the Legendre symbol of  $a$  and  $p$ . Indeed, multiplication by  $a$  is a bijection on  $\mathbb{F}_p^*$  which permutes the subsets  $\{x_i, -x_i\}$ . (We defined  $(\frac{b}{p})$  for  $b \in \mathbb{Z}$  prime to  $p$ , but it is clear that, once  $p$  is chosen,  $(\frac{b}{p})$  depends only on the congruence class  $\bar{b}$  of  $b \pmod{p}$ , so that  $(\frac{\bar{b}}{p})$  is well-defined. Also we regard  $(\frac{c}{p})$  as equal to  $\pm 1$  sometimes in  $\mathbb{Z}$ , sometimes in  $\mathbb{F}_p$ .)

**Theorem 10** (Gauss's Lemma, strong form). *In the above setting we have*

$$\left(\frac{a}{p}\right) = \prod_{i=1}^n \frac{ax_i}{x_{\sigma(i)}}. \quad (12)$$

*First proof of Theorem 10.* Since the right-hand side is equal to  $a^n$ , the result follows from Euler's Criterion (Theorem 6).  $\square$

The above proof is especially short, but slightly mysterious. The following proof will be a little bit longer, but hopefully less mysterious. More precisely, we hope the next proof will make clearer where the expression in the right-hand side of (12) comes from.

**Lemma 11.** *The Legendre symbol  $(\frac{a}{p})$  is the sign of the permutation  $x \mapsto ax$  of  $\mathbb{F}_p^*$ .*

*Proof.* Let  $\zeta$  be a generator of  $\mathbb{F}_p^*$  (see Theorem 8). It suffices to show that the permutation  $x \mapsto \zeta x$  of  $\mathbb{F}_p^*$  is odd (see Corollary 7) (indeed, both  $x \mapsto (\frac{x}{p})$  and the sign of the permutation  $y \mapsto xy$  of  $\mathbb{F}_p^*$  are multiplicative homomorphisms  $\mathbb{F}_p^* \rightarrow \{\pm 1\}$ , so it suffices to check they agree on a generator). But this permutation is the  $2n$ -cycle  $(\zeta^1, \zeta^2, \dots, \zeta^{2n})$ .  $\square$

**Theorem 12.** *Let  $\sigma$  be a permutation of  $\mathbb{F}_p^*$  satisfying  $\sigma(-x) = -\sigma(x)$  for all  $x \in \mathbb{F}_p^*$ . Then there is a unique permutation  $\sigma'$  of  $I = \{1, \dots, n\}$  such that  $\sigma(x_i) = \pm x_{\sigma'(i)}$  for all  $i \in I$ , and we have*

$$\text{sgn}(\sigma) = \prod_{i=1}^n \frac{\sigma(x_i)}{x_{\sigma'(i)}}.$$

*Proof.* The existence of  $\sigma'$  is clear. Let  $K$  be the set of those  $i \in I$  such that  $\sigma(x_i) = -x_{\sigma'(i)}$ . We must show that the sign of  $\sigma$  is  $(-1)^{|K|}$ , where  $|K|$  is the cardinality of  $K$ . It suffices to define permutations  $\sigma_1, \dots, \sigma_m$  of  $\mathbb{F}_p^*$  such that:

$$\sigma = \sigma_m \circ \dots \circ \sigma_1, \quad m = |K| + 2, \quad (13)$$

$$\text{sgn}(\sigma_1) = \text{sgn}(\sigma') = \text{sgn}(\sigma_2), \quad \text{sgn}(\sigma_3) = \text{sgn}(\sigma_4) = \dots = \text{sgn}(\sigma_m) = -1. \quad (14)$$

To do that, we set

$$\sigma_1(x_i) = x_{\sigma'(i)}, \quad \sigma_1(-x_i) = -x_i, \quad \sigma_2(x_i) = x_i, \quad \sigma_2(-x_i) = -x_{\sigma'(i)},$$

and we define  $\sigma_3, \sigma_4, \dots, \sigma_m$  as follows:

$\sigma_3$  is the transposition  $(x_{\sigma'(k)}, -x_{\sigma'(k)})$  where  $k$  is the least element of  $K$ ,

$\sigma_4$  is the transposition  $(x_{\sigma'(k)}, -x_{\sigma'(k)})$  where  $k$  is the next element of  $K$ ,

$\dots$ ,

$\sigma_m$  is the transposition  $(x_{\sigma'(k)}, -x_{\sigma'(k)})$  where  $k$  is the largest element of  $K$ .

In particular  $\sigma_2 \circ \sigma_1$  sends  $x_i$  to  $x_{\sigma'(i)}$  and  $-x_i$  to  $-x_{\sigma'(i)}$ . We have  $\text{sgn}(\sigma_1) \cdot \text{sgn}(\sigma_2) = \text{sgn}(\sigma')^2 = 1$ , and each of  $\sigma_3, \dots, \sigma_m$  is a transposition, giving  $\text{sgn}(\sigma) = (-1)^{|K|}$ , and we easily check the equalities (13) and (14).  $\square$

*Second proof of Theorem 10.* Theorem 10 follows from Lemma 11 and Theorem 12.  $\square$

## 5 A generalization of Euler's Criterion

Let  $m$  be a positive integer and  $G$  the group  $\mathbb{Z}/m\mathbb{Z}$ . For each integer  $k$  we set  $kG = \{k\gamma \mid \gamma \in G\}$  and we define  $f_k : G \rightarrow G$  by  $f_k(\gamma) = k\gamma$ . Note that  $kG$  is a subgroup of  $G$  and  $f_k$  is an endomorphism of  $G$ .

**Proposition 13.** *If  $k$  is an integer and if  $d = \gcd(k, m)$ , then  $\text{Im } f_k = kG = dG = \text{Ker } f_{m/d}$ , and the order of this subgroup is  $\frac{m}{d}$ .*

We get Euler's Criterion by setting  $m = p - 1$  and  $k = 2$ , which gives  $d = 2$  and  $\frac{m}{d} = (p - 1)/2$ .

*Proof.* To prove  $kG = dG$ , note that we have  $k\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$  by definition of  $d$ , and that these subgroups of  $\mathbb{Z}$  project respectively to  $kG$  and to  $dG$  in  $G$ . To show  $dG \subset \text{Ker } f_{m/d}$  note that  $\frac{m}{d}d\gamma = m\gamma = 0$  for  $\gamma \in G$ . Let us check  $\text{Ker } f_{m/d} \subset dG$ . Let  $x$  be an integer such that  $\frac{m}{d}x \equiv 0 \pmod{m}$ . We must show that  $d$  divides  $x$ . But we have  $\frac{m}{d}x \in m\mathbb{Z}$ , and thus  $\frac{x}{d} \in \mathbb{Z}$ . It only remains to prove  $|dG| = \frac{m}{d}$ . Setting  $\bar{x} = x \bmod m$  for  $x \in \mathbb{Z}$ , we get, since  $\frac{m}{d}\bar{d} = \bar{0}$ ,

$$dG = \{\dots, -2\bar{d}, -\bar{d}, \bar{0}, \bar{d}, 2\bar{d}, \dots\} = \{\bar{0}, \bar{d}, 2\bar{d}, \dots, (\frac{m}{d} - 1)\bar{d}\},$$

and the elements  $\bar{0}, \bar{d}, 2\bar{d}, \dots, (\frac{m}{d} - 1)\bar{d}$  are distinct because  $0 < d < 2d < \dots < (\frac{m}{d} - 1)d < m$ .  $\square$

## References

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